

School of Engineering and Technology

Department of Electrical and Electronics Engineering

LOG BOOK

Academic Year: 2026-27

Semester: 7th

Type of Project: Capstone Project

Course code:

Group Number: ~~25266 pg-86~~ 262757-121

Title of the Project: Vision-Based Autonomous Pick-and-Place Robotic Arm Using Imitation Learning and Digital Twin

Aim: To transform a 6-DOF robotic arm into an autonomous, vision-guided pick-and-place manipulation system that learns behaviour directly from human teleoperated demonstrations using imitation learning, an OpenCV-based perception module, and a digital twin of the arm and workspace.

Objectives:

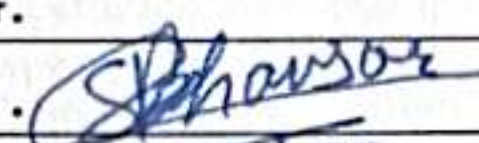
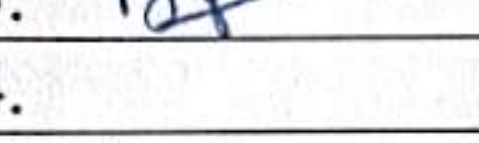
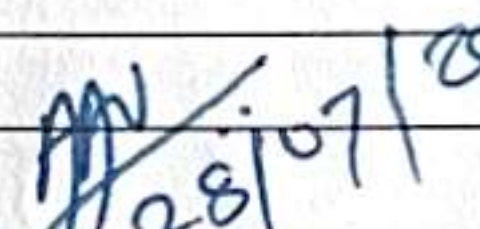
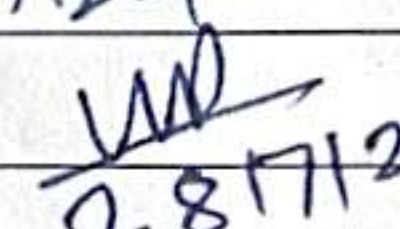
1. Develop an imitation learning-based robotic manipulation system.
2. Develop a PS5 controller-based teleoperation system utilizing Inverse Kinematics for smooth demonstration collection.
3. Build a synchronized dual-camera data acquisition framework.
4. Develop a robust OpenCV computer vision module for workspace perception (utilizing ArUco markers and color spaces).
5. Train an imitation learning model capable of generating full 6-DOF robotic arm trajectories.
6. Develop a digital twin of the robotic arm and workspace.
7. Deploy the trained model on a Raspberry Pi 5.
8. Evaluate the system under progressively increasing task complexity.

Name of the Guide: Prof. Aarti Vagga

Name of the Co-Guide: Dr. Vaidehi Deshmukh

July 2026		
Sr. No.	Details of the Tasks	Description
1.	Date & Duration:	Date of Meeting: July 20 to 31, 2026
		Total Hours Spent on Project During the Period:
Name of the students	Individual Task assigned	Individual Task completion
Soham Bhavsar	Finalize hardware architecture, mechanically optimize joint limits, integrate the 6-DOF arm with Arduino Uno, calibrate the PCA9685 PWM driver, develop the Inverse Kinematics solver and PS5 controller interface, install the dual-camera system, and define the Home Position.	Dual-camera setup initiated. IK solver initiated. Built a complete dashboard to help the complete operations of the project throughout.
Toshali	Begin the CAD model of	Finalized the CAD model of the 6-DOF

Kumbhar	the 6-DOF serial arm and start digital twin/workspace modelling, including planning camera placement within the simulation environment.	robotic arm. Initiated motion simulation trials within the digital twin environment.
Divyansh Dewangan	Assist with testing the PS5 teleoperation interface for smoothness of Cartesian commands and help document the Home Position and motion constraints for consistent demonstration episodes.	Drafted the demonstration-collection methodology. Conducted initial teleoperation trials to assess data quality ahead of upcoming dataset collection.
2.	Which of the stated objectives is achieved?	Objective 2 (PS5 controller-based teleoperation with IK) and Objective 3 (dual-camera framework) initiated; Objective 6 (digital twin) begun.
3.	Detailed summary of tasks completed	Software used: Arduino IDE (firmware for PCA9685/servo control), Python (Inverse Kinematics solver and PS5 controller interface), CAD/simulation software for the digital twin. Hardware used: 6-DOF robotic arm (3 x MG996R, 3 x MG90S), Arduino Uno, PCA9685 16-channel PWM driver, PlayStation 5 controller, two fixed USB cameras, physical Emergency Stop switch. Experiments conducted: Bench testing of joint limits and Home Position; validation of IK-based Cartesian-to-joint mapping using the PS5 controller. Outcome of the experiment: Built a project dashboard and a project journal to help track progress and manage the project throughout its duration. List of performance parameters used for analysis: Teleoperation loop latency, accuracy of IK-computed joint angles against commanded end-effector position.
4.	Challenges Faced	Technical or non-technical issues encountered: Steps taken to address them
5.	Key learning:	Concepts understood: Inverse Kinematics for serial manipulators; PWM-based servo control over I2C. Skills gained: Arduino-PCA9685 hardware interfacing; real-time teleoperation control design.
6.	Inputs from the Guide:	Feedback:

		Suggestions:
7.	Action Plan for Next Meeting	1. Develop the OpenCV perception module with ArUco tracking for Stage 1.
		2. Detect object position and orientation in real-world coordinates.
		3. Synchronize camera feeds with the teleoperation loop.
		4. Develop the dataset recording framework.
		5. Validate synchronized recording integrity.
		6. Continue digital twin development.
8.	Resources Used:	Research Papers: Literature on Inverse Kinematics solutions for 6-DOF serial manipulators.
		Books:
		Datasets: Not applicable this period (demonstration collection has not yet begun).
9.	Percentages of Actual work completed	
10.	Individual Contributions	1. Soham Bhavsar
		2. Toshal Kumbhar
		3. Divyansh Dewangan
		4.
11.	Signature of candidate with date	1. 
		2. 
		3. 
		4.
		Guide Signature with Date:  28/07/2026
		Co-Guide Signature with Date:  28/7/26